



King Saud University

College of Computer and Information Sciences

Department of Computer Science

CSC 227: Operating Systems Programming Project 1: Process Scheduling Simulator

First Semester 1447

Student Name	Student ID
لين الزهراني	444200939
رذاذ بن راكان	444200451
رزان المطيري	444200495
منيره الحويكان	444200541
سارة الصبيحي	444200926

Task Distribution Details

لين الزهراني – Project Setup & Input Handling :

- Created the Process class.
- Handled user input for:
 - Number of processes
 - Arrival time, burst time, and priority
 - Time quantum (for Round Robin)
- Initialized data structures (ArrayList, etc.).
- Added input validation (no negative times).

رذاذ بن راكان – Preemptive Priority + Round Robin Logic:

- Implemented Preemptive Priority Scheduling algorithm.
- Managed Round Robin scheduling for processes with the same priority.
- Ensured preemption occurs correctly when a higher-priority process arrives.
- Debugged and verified correct execution order.

رزان المطيري – FCFS Scheduling + Metrics :

- Implemented First-Come, First-Served (FCFS) scheduling algorithm.
- Calculated:
 - Average Turnaround Time
 - Average Waiting Time
 - CPU Utilization.
- Ensured metric calculations are correct for multiple test cases.

منيره الحويكان – Output Visualization & Report :

- Developed clear Gantt Charts for both algorithms (FCFS and Priority+RR).
- Organized and formatted the final console outputs.
- Created this ReadMe Report summarizing the project and teamwork.

ساره الصبيحي – Testing & Additional Run Cases:

- display outputs clearly, collect and compare outputs for both algorithms.
- Designed and executed extra test cases, including:
 - Case A: Different priorities and short burst times.
 - Case B: Same priority (to test Round Robin).
 - Case C: Late-arriving high-priority process (to test preemption).
- Verified accuracy of results.

Screenshots and Outputs

Priority Queue with Round Robin Output

- Shows the correct handling of process preemption and time quantum rotation for same-priority tasks.
- Displays calculated metrics and CPU utilization.

```
All processes entered:
P1 arrival: 0 burst: 12 priority: 1
P2 arrival: 1 burst: 3 priority: 1
P3 arrival: 2 burst: 6 priority: 2

==== Priority Queue with Round Robin ====

Gantt Chart:
-----
| P1 | P2 | P1 | P2 | P1 | P1 | P1 | P1 | P3 | P3 | P3 |
-----
0   2   4   6   7   9  11  13  15  17  19  21

Time    Process
0-2     P1
2-4     P2
4-6     P1
6-7     P2
7-9     P1
9-11    P1
11-13   P1
13-15   P3
15-17   P3
17-19   P3
19-21   P3

Performance Metrics
Average Turnaround Time: 13.33
Average Waiting Time: 6.33
CPU Utilization: 100%
```

Figure 1 Output of Priority + Round Robin Scheduling

FCFS Scheduling Output

- Demonstrates sequential execution based on arrival time.
- Displays Gantt chart, turnaround, and waiting time metrics.

```
==== First-Come First-Served (FCFS) ====

Time    Process
0-12    P1
12-15   P2
15-21   P3

Gantt Chart:
-----
| P1 | P2 | P3 |
-----
0    12   15   21

Performance Metrics
Average Turnaround Time: 15.00
Average Waiting Time: 8.00
CPU Utilization: 100%
```

Figure 2: Output of First-Come, First-Served (FCFS) Scheduling

Testing

Case A – Different Priorities, Short Bursts

Purpose:

Test how the scheduler handles multiple priorities when processes have short CPU bursts.

Input:

Number of processes: 3

P1 → Arrival: 0, Burst: 4, Priority: 3

P2 → Arrival: 1, Burst: 3, Priority: 1

P3 → Arrival: 2, Burst: 2, Priority: 2

Quantum: 2

Output Screenshots:

Priority + RR:

```
All processes entered:
P1 arrival: 0 burst: 4 priority: 3
P2 arrival: 1 burst: 3 priority: 1
P3 arrival: 2 burst: 2 priority: 2

==== Priority Queue with Round Robin ====

Gantt Chart:
-----
| P1 | P2 | P2 | P3 | P1 |
-----
0   2   4   5   7   9

Time   Process
0-2    P1
2-4    P2
4-5    P2
5-7    P3
7-9    P1

Performance Metrics
Average Turnaround Time: 6.00
Average Waiting Time: 3.00
CPU Utilization: 100%
```

FCFS:

```
==== First-Come First-Served (FCFS) ====

Time   Process
0-4    P1
4-7    P2
7-9    P3

Gantt Chart:
-----
| P1 | P2 | P3 |
-----
0     4     7     9

Performance Metrics
Average Turnaround Time: 5.67
Average Waiting Time: 2.67
CPU Utilization: 100%
```

Observations:

- Priority + RR gives the CPU to higher-priority processes as soon as the current time slice finishes. After P1 finishes its first quantum (0–2), P2 (higher priority) is scheduled before lower-priority work.
- FCFS executes strictly in arrival order, delaying high-priority tasks.

Case B – Same Priority (Round Robin Test)

Purpose:

Verify that Round Robin is correctly applied when all processes have the same priority.

Input:

Number of processes: 4

P1 → Arrival: 0, Burst: 5, Priority: 1

P2 → Arrival: 1, Burst: 3, Priority: 1

P3 → Arrival: 2, Burst: 4, Priority: 1

P4 → Arrival: 3, Burst: 2, Priority: 1

Quantum: 2

Output Screenshots:

Priority + RR:

```
===== Priority Queue with Round Robin =====

Gantt Chart:
-----
| P1 | P2 | P3 | P1 | P4 | P2 | P3 | P1 |
-----
0   2   4   6   8   10  11  13  14

Time    Process
0-2     P1
2-4     P2
4-6     P3
6-8     P1
8-10    P4
10-11   P2
11-13   P3
13-14   P1

Performance Metrics
Average Turnaround Time: 10.50
Average Waiting Time: 7.00
CPU Utilization: 100%
```

FCFS:

```
===== First-Come First-Served (FCFS) =====

Time    Process
0-5     P1
5-8     P2
8-12    P3
12-14   P4

Gantt Chart:
-----
| P1 | P2 | P3 | P4 |
-----
0    5    8   12  14

Performance Metrics
Average Turnaround Time: 8.25
Average Waiting Time: 4.75
CPU Utilization: 100%
```

Observations:

- Priority + RR distributes CPU time fairly among all same-priority processes.

- FCFS favors earlier arrivals, causing later processes to wait longer

Case C – Late-Arriving High-Priority Process (Preemption Test)

Purpose:

Check whether the scheduler correctly preempts running processes when a higher-priority process arrives late.

Input:

Number of processes: 4

P1 → Arrival: 0, Burst: 6, Priority: 3

P2 → Arrival: 1, Burst: 4, Priority: 2

P3 → Arrival: 4, Burst: 5, Priority: 1

P4 → Arrival: 5, Burst: 2, Priority: 2

Quantum: 2

Output Screenshots:

Priority + RR:

```
===== Priority Queue with Round Robin =====
Gantt Chart:
-----
| P1 | P2 | P3 | P3 | P3 | P2 | P4 | P1 | P1 |
-----
0   2   4   6   8   9  11  13  15  17

Time   Process
0-2    P1
2-4    P2
4-6    P3
6-8    P3
8-9    P3
9-11   P2
11-13  P4
13-15  P1
15-17  P1

Performance Metrics
Average Turnaround Time: 10.00
Average Waiting Time: 5.75
CPU Utilization: 100%
```

FCFS:

```
===== First-Come First-Served (FCFS) =====
Time   Process
0-6    P1
6-10   P2
10-15  P3
15-17  P4

Gantt Chart:
-----
| P1 | P2 | P3 | P4 |
-----
0    6   10  15  17

Performance Metrics
Average Turnaround Time: 9.50
Average Waiting Time: 5.25
CPU Utilization: 100%
```

- **Observations:**
- Priority + RR preempts lower-priority tasks (P1, P2) when P3 arrives.
- FCFS cannot preempt; high-priority tasks wait until their turn.
- Demonstrates the advantage of preemptive scheduling for urgent tasks.

Summary of Testing

- **Priority + RR** improves responsiveness for high-priority tasks and ensures fairness among equal-priority processes.
- **FCFS** is simpler but can delay critical tasks if a high-priority process arrives late.
- All algorithms were **validated using multiple test cases**, and outputs match expected Gantt charts and performance metrics.